

CS 466: Practice Final Problems

Problem 1. Define $G: \{0, 1\}^{128} \times \{0, 1\}^{128} \rightarrow \{0, 1\}^{128}$ by

$$G_K(x) = \text{AES}_K(x) \oplus \text{AES}_K(x \oplus 1^{128}).$$

Show that G is *not* a secure PRF.

Problem 2. Let $\text{SE} = (\mathcal{K}, \mathcal{E}, \mathcal{D})$ have key space $\{0, 1\}^{128}$ and message space $\{0, 1\}^{256}$. Encryption samples $r \leftarrow_{\$} \{0, 1\}^{128}$ and sets:

$$\mathcal{E}_K(M_1 \| M_2) = r \parallel (\text{AES}_K(r) \oplus M_1) \parallel (\text{AES}_K(M_1) \oplus M_2).$$

Show that SE is *not* IND-CPA secure.

Problem 3. Define $H: \{0, 1\}^{128} \times \{0, 1\}^{256} \rightarrow \{0, 1\}^{128}$ by

$$H_K(M_1 \| M_2) = \text{AES}_K(M_1) \oplus M_2, \quad M_1, M_2 \in \{0, 1\}^{128}.$$

Show that H is *not* collision resistant.

Problem 4. Let D be the set of all bit strings of length a positive multiple of 128. Let $\text{CBCMAC}_K(M)$ denote plain CBC-MAC with key K . To prevent splicing attacks, Alice proposes the following modification: for $M = M[1] \parallel \dots \parallel M[m] \in D$, define

$$\mathcal{T}_K(M) = \text{CBCMAC}_K(M) \oplus M[1] \oplus \dots \oplus M[m].$$

Show that $\mathcal{T}$ is *not* UF-CMA secure.

Problem 5. Let $\text{AE} = (\mathcal{K}, \mathcal{E}, \mathcal{D})$ have key space $\{0, 1\}^{128}$ and message space $\{0, 1\}^{128}$. Encryption samples $r \leftarrow_{\$} \{0, 1\}^{128}$ and computes $C = \text{AES}_K(r) \oplus M$, then sets:

$$\mathcal{E}_K(M) = r \parallel C \parallel \text{AES}_K(r \oplus C).$$

Decryption of (r, C, T) returns $\text{AES}_K(r) \oplus C$ if $\text{AES}_K(r \oplus C) = T$, and $\perp$ otherwise.

Show that AE is *not* INT-CTXT secure.

Problem 6. Let $h: \{0, 1\}^{128} \times \{0, 1\}^{128} \rightarrow \{0, 1\}^{128}$ be a keyless collision-resistant compression function. Consider the variant of MD in which the initial chaining value is set to the *first message block* and the remaining blocks are processed by the usual iteration, with no MD-strengthening (no padding or length-encoding) applied. Let $H: D \rightarrow \{0, 1\}^{128}$ denote this variant built from h , where D is the set of bit strings of positive length a multiple of 128.

Part A. Show that H is *not* collision resistant.

Part B. Can you find an attack when H is modified to add back MD-strengthening? Discuss.

Problem 7. A *stateful* symmetric encryption scheme is a triple $\text{SE} = (\mathcal{K}, \mathcal{E}, \mathcal{D})$ in which key generation outputs both a key K and an initial state σ_0 , and the encryption algorithm takes the current state σ as additional input alongside the message M and returns a ciphertext C together with an updated state σ' ; we write $(C, \sigma') \leftarrow \mathcal{E}_K(\sigma, M)$. The IND-CPA game in this model samples $(K, \sigma_0) \leftarrow \mathcal{K}$ and on each $\text{LR}(M_0, M_1)$ query computes $(C, \sigma') \leftarrow \mathcal{E}_K(\sigma, M_b)$ —where $b = 1$ in the right game and $b = 0$ in the left game—updates $\sigma \leftarrow \sigma'$, and returns C . Advantage is defined as usual. Consider the scheme $\text{SE} = (\mathcal{K}, \mathcal{E}, \mathcal{D})$ with key space $\{0, 1\}^{128}$ and message space $\{0, 1\}^{128}$:

$$\begin{aligned} \mathcal{K}: & \quad K \leftarrow \mathcal{K}; \sigma_0 \leftarrow \mathcal{K}; \text{ return } (K, \sigma_0) \\ \mathcal{E}_K(\sigma, M): & \quad C \leftarrow \text{AES}_K(\sigma \oplus M); \text{ return } ((\sigma, C), C) \end{aligned}$$

That is, a ciphertext is the pair (σ, C) (the IV used together with the encrypted block), and the state then advances to C . For context, this is essentially how SSL/TLS 1.0 chained CBC records.

Part A. Show that SE is *not* IND-CPA secure.

Part B. Define a stateful variant of CTR\$-encryption that is IND-CPA in the above sense, yet its encryption algorithm does not flip coins.

Problem 8. Let CBCMAC_K denote basic CBC-MAC built from AES_K on the set of bit strings of positive length a multiple of 128. Define

$$\mathcal{T}_K(M) = \text{CBCMAC}_K(M \parallel \text{CBCMAC}_K(M)),$$

i.e., compute the inner CBC-MAC tag of M , append it to M , and CBC-MAC the result.

Part A. Show that $\mathcal{T}$ is *not* UF-CMA secure. *Hint:* what happens when M is a single block?

Part B. Can you find an attack on $\hat{\mathcal{T}}_K(M) = \text{CBCMAC}_K(\text{CBCMAC}_K(M) \parallel M)$? Discuss.